

1. What is the output of the following code snippet?

```
int a = 5, b = 10;
cout << (a > b) << endl;
```

2. Evaluate the expression: `!(5 > 3 && 2 < 4)`

3. Given `int x = 7, y = 3;`, what is the result of `(x != y) || (x < y)`?

4. Write a C++ expression that checks if a variable `num` is between 0 and 100 (inclusive) using logical operators.

5. What is the difference between the relational operator `==` and the assignment operator `=`? Provide an example of a common mistake.

6. Predict the output:

```
bool flag = false;
cout << (flag && (5 > 2)) << endl;
```

7. Why does the expression `(3 < 5 < 2)` produce unexpected results in C++? Explain how the operators associate.

8. Determine the truth value of: `(10 >= 10) && (5 != 6) || (2 > 3)`

9. What will the following code print?

```
int a = 0, b = 1;
if (a++ && b++)
    cout << "True";
else
    cout << "False";
cout << a << b;
```

10. Write a logical expression that returns `true` if a character `ch` is a vowel (a, e, i, o, u) regardless of case.

11. Explain the short-circuit behavior of `&&` and `||` operators. How does it affect the following code?

```
int x = 0;
if (x != 0 && 10 / x > 2)
    cout << "Safe";
else
    cout << "Not safe";
```

12. Find the error in this code:

```
int age = 18;
if (age => 18)
    cout << "Adult";
```

13. Evaluate: `(true || false && true)`. Show the order of evaluation.

14. Write a single C++ condition that checks if a number `n` is not in the range `[10, 20]` (i.e., less than 10 or greater than 20).

15. What is the output of the following?

```
cout << (3 > 2 > 1) << endl;
```

16. Given `int p = 5, q = 5;`, what is `(p == q) != (p = q)`? Explain.

17. Construct a logical expression that is `true` only when exactly two of three boolean variables `a`, `b`, `c` are `true`.

18. What will this code print?

```
int i = 1, j = 2;
if (i == 1 || j++ == 3)
    cout << j;
```

19. Why does `(5 > 4) == 1` evaluate to `true` in C++? What does the relational operator return?

20. Write a condition using relational and logical operators that checks if a year `year` is a leap year (divisible by 4 but not by 100, unless also divisible by 400).